8. Assign four trucks 1, 2, 3 and 4 to vacant spaces A, B, C, D, E and F so that distance travelled is minimized. The matrix below shows the distance.

	1	2	3	4
A	4	7	3	7
B	8	2	5	5
C	4	9	6	9
D	7	5	4	8
E	6	3	5	4
F	6	8	7	3

10. 3 Decision Theory

Introduction:

Decision theory is primarily concerned with helping people and organizations in making decisions. It provides a meaningful conceptual frame work for important decision making. The decision making refers to the selection of an act from amongst various alternatives, the one which is judged to be the best under given circumstances.

The management has to consider phases like planning, organization, direction, command and control. While performing so many activities, the management has to face many situations from which the best choice is to be taken. This choice making is technically termed as "decision making" or decision taking. A decision is simply a selection from two or more courses of action. Decision making may be defined as "a process of best selection from a set of alternative courses of action, that course of action which is supposed to meet objectives upto satisfaction of the decision maker."

The knowledge of statistical techniques helps to select the best action. The statistical decision theory refers to an optimal choice under condition of uncertainty. In this case probability theory has a vital role, as such, this probability theory will be used more frequently in the decision making theory under uncertainty and risk.

The statistical decision theory tries to reveal the logical structure of the problem into alternative action, states of nature, possible outcomes and likely pay-offs from each such outcome. Let us explain the concepts associated with the decision theory approach to problem solving.

10.3.1 Meaning

The decision maker: The decision maker refers to individual or a group of individual responsible for making the choice of an appropriate course of action amongst the available courses of action.

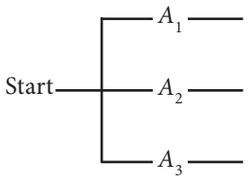
Acts (or courses of action): Decision making problems deals with the selection of a single act from a set of alternative acts. If two or more alternative courses of action occur in a problem, then decision making is necessary to select only one course of action. Let the acts or action be a_1, a_2, a_3,... then the totality of all these actions is known as action space denoted by A. For three actions a_1, a_2 a_3; A = action space = (a_1, a_2, a_3) or A = (A_1, A_2, A_3). Acts may be also represented in the following matrix form.

Acts
A_1
A_2
...
A_n

(or)

Acts	A_1	A_2	...	A_n

In a tree diagram the acts or action are shown as



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